Fact-Grounded Small Language Models and Code-Mixed Clinical NER for Privacy-Preserving On-Device Explanation of Bengali Health Reports

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***Abstract*—Bengali speakers routinely receive laboratory reports and prescriptions written in English or in code-mixed Bengali–English ("Banglish"), and increasingly paste them into cloud chatbots, exposing personal health information. We study whether small, locally runnable language models can explain such reports safely in Bengali, and build ShasthoSathi, an offline mobile pipeline with four stages: hybrid de-identification, code-mixed medicine recognition, rule-based abnormality flagging, and explanation. On held-out synthetic reports and on real clinical text with injected identifiers, combining format rules with OpenMed PII models and protecting fine-tuned medicine spans reduced the personal-detail leak rate from 10.3% (PII models alone) to 0% (95% upper bound 0.16%) while limiting clinical over-redaction to 6.9% on real text. Fine-tuning on Bangla-HealthNER raised Bengali-script medicine recall from below 10% (zero-shot) to 81%, and INT8 export reduced the tagger to 106 MB with a 0.09-point F1 loss. Four instruction-tuned models of 0.5B–3.8B parameters flipped high/low in 17–42% of abnormal results when reading reports directly; a rule-computed fact sheet and a strict no-interpretation prompt raised Qwen3-1.7B to 97% correct and 96% safe in English, as judged by a validated LLM judge (Cohen's $\kappa = 0.80$). In Bengali, every model failed: a blinded native-speaker check rated a deterministic template 100% correct versus 60% for the best model. We further show that standard dynamic INT8 quantization silently destroys a Bengali PII model's ability to find unlabelled names (100% missed), whereas quantizing only the embedding table preserves full accuracy at half the size. On a Galaxy S23 Ultra, the full pipeline ran offline in a median 262 ms per report and matched the server's leak rate on 1,100 test documents, but only after we found that INT8 matrix multiplications made the English PII model miss 15.5% of free-text names on the phone versus 1.0% on the server. The results argue for template-first generation with LLMs as optional rewriters in low-resource languages. Project page, code and an in-browser demo: https://cimilab.github.io/Bangla_ShasthoSathi/.**

***Index Terms*—Bengali NLP, code-mixed named entity recognition, de-identification, small language models, grounded generation, LLM-as-a-judge, quantization, on-device inference, patient health literacy**

I. Introduction

Laboratory reports are written for clinicians. Patients find them hard to interpret and are often left to infer what an out-of-range value means [1]. Large language models (LLMs) can rewrite clinical text in plain language [2] and answer lay questions about laboratory results [3], but two obstacles block their use for Bengali speakers. First, general chatbots run in the cloud, so a photographed report sends names, phone numbers, national ID numbers and diagnoses to a third party. Second, Bengali is a low-resource language for LLMs: tokenizers split Bengali text into many more tokens than English [4], and small models that fit on a phone have seen little Bengali medical text.

Bangladeshi health documents add a third difficulty: they are code-mixed. In the Bangla-HealthNER corpus [5], 60.5% of medicine mentions are written in Latin script, 37.6% in Bengali script, and dosage patterns such as "1+0+1" mix both. An English clinical NER model therefore misses Bengali-script drug names, and a Bengali PII model can erase the very drug names a patient needs explained.

This paper asks whether a privacy-first, offline pipeline built from small models can explain such documents correctly, and where small LLMs help or harm. We make five contributions:

1) A **hybrid de-identification** method that unions format rules with OpenMed PII models [6], routed by script, and subtracts fine-tuned medicine and dosage spans so that clinical content survives (Sec. V-B).
2) A **code-mixed medicine tagger** fine-tuned on Bangla-HealthNER and exported to a 106 MB INT8 model for mobile CPUs (Sec. V-C).
3) A controlled study of **grounded generation** with four small LLMs in English and Bengali, evaluated by an LLM judge whose rubric was validated before use and by a blinded native-speaker check (Sec. V-D).
4) A finding on **quantization fragility**: dynamic INT8 quantization disables unlabelled-name detection in a multilingual Bengali PII model while leaving an English model intact; embedding-only quantization avoids the loss, and INT8 matrix multiplications make an English PII model's errors hardware-dependent (15.5% vs. 1.0% of free-text names missed on the phone's ARM CPU and the x86 server) (Sec. VI-D).
5) An Android implementation whose Kotlin core is verified byte-for-byte against the Python reference on 909 texts, with on-device OCR and an evaluation protocol

for photographed reports (Sec. VI-D).

II. Related Work

Clinical de-identification. Shared tasks established recurrent and transformer taggers for protected health information in English notes [7], [8], and rule-plus-model systems such as Philter favor recall for privacy [9]. OpenMed released open PII and clinical NER transformers, including a Bengali PII model [6]. We extend the hybrid idea to code-mixed Bengali and add clinical-entity protection, which earlier systems did not need because they rarely de-identify text that the patient must still read.

Bengali NLP and medical NER. BanglaBERT and BanglishBERT are ELECTRA-style encoders for Bengali and Bengali–English text [10]; XLM-RoBERTa covers 100 languages [11]. Bangla-HealthNER provides 31,783 annotated health posts with seven entity types and reports 56.13 micro-F1 for BanglishBERT [5].

LLMs for patient-facing explanation. LLMs simplify radiology reports with occasional harmful errors [2] and answer laboratory questions better than peer users but with hallucinations [3], [12]. Med-PaLM introduced human evaluation axes such as factuality, harm and bias [13]; PEMAT measures understandability of patient materials [14]. LLM judges agree well with humans on many English tasks [15], but their reliability in low-resource languages is less studied; we validate ours before use and compare it with a native speaker.

On-device models. Small models such as Qwen [16], [17] and Phi-3 [18] target local deployment; 4-bit NF4 loading [19] and integer quantization [20] reduce memory. Tesseract [21] and ML Kit [22] provide offline OCR.

III. System Overview

Figure 1 shows the pipeline. Two design rules follow from our experiments: (i) *an LLM never decides whether a value is abnormal*; rules compare each value with the report's own reference range, and (ii) *an LLM never sees the report*; it receives only a rule-computed fact sheet after de-identification. All stages run offline; the Android app declares no internet permission, so the operating system itself prevents data from leaving the phone.

IV. Data

No licensed or data-use-agreement corpora were used. Table I lists the sets.

Bangla-HealthNER [5]: 25,426 / 3,178 / 3,179 train/validation/test posts with Medicine, Dosage, Health Condition, Symptom, Specialist, Age and Medical Procedure entities, used for tagger training (E4) and as real Bengali text.

Synthetic reports. Gold-offset documents generated from templates of Bangladeshi lab reports (12 tests with reference ranges), mixed and English prescriptions (common brand names with generic, strength, dose pattern, food relation and duration) and Bengali clinical notes, with invented names, phone numbers, national IDs, addresses, facilities and dates. A development set (300 documents) uses one vocabulary of names, places and facilities; a held-out set (300 documents; 1,892 identifier spans, 452 lab rows) uses a disjoint vocabulary, so rules cannot memorize names.

Real text with injected identifiers. 200 PMC-Patients summaries [23] and 200 Bangla-HealthNER test posts, each wrapped with a fake patient name, phone, date, doctor and facility at known offsets (2,000 spans).

Free-text names. Because injected identifiers follow field labels ("Patient:", "Dr."), rules can catch them by format. We therefore built a harder set in which the same real texts contain one name inside a sentence without any label ("… was admitted", Bengali "my father … has been ill"): a test set (400) and a disjoint development set (400) for choosing settings.

Printed sheets (E8). 30 held-out reports (15 lab reports, 8 mixed prescriptions, 7 Bengali notes) rendered to A4 for photographing.

A. Preprocessing and cleaning

Word segmentation and label alignment. Bangla-HealthNER posts are split on whitespace, giving one BIO tag per word. Each word is tokenized separately; the tag is assigned to its first sub-word and the remaining sub-words are ignored in the loss (label -100). Training sequences are truncated at 256 sub-words; at test time a word cut off by truncation counts as O, so truncation can only lower our scores. On the phone, long texts are instead processed in overlapping windows of 254 sub-words with a 64-token stride, and a token takes its prediction from the first window that contains it.

Script profiling (E0). Every gold mention is classified as Latin, Bengali or mixed script from its Unicode blocks (U+0980–U+09FF for Bengali). Fig. 2 shows why a monolingual model cannot suffice: medicines are mostly Latin, symptoms mostly Bengali, and a quarter of dosages mix both.

Numeric normalization. Bengali digits (U+09E6–U+09EF) are mapped to ASCII before parsing, decimal commas are accepted, and Unicode whitespace variants are collapsed. The same normalization is applied in the gate, so a Bengali-digit number in an LLM output is matched against the facts.

Span cleaning. Model spans are merged when they overlap or touch, protected spans are subtracted, and the result is trimmed of surrounding whitespace and punctuation. Trimming never removes letters or Bengali vowel signs (dependent combining marks), which would otherwise leave a visible fragment of a name.

Synthetic gold integrity. Generated documents record each identifier's character offsets at creation time, and the generator asserts that every offset range reproduces the span's surface string. For OCR scoring, reference and OCR text are whitespace-normalized before the character error rate is computed.

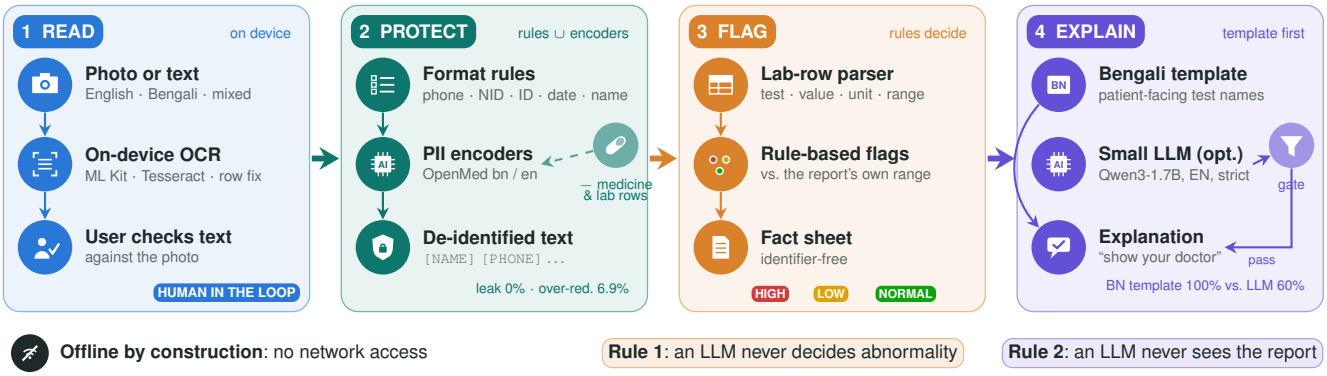


Fig. 1. ShasthoSathi pipeline. **(1) Read**: on-device OCR, then the user checks the text against the photo. **(2) Protect**: format rules ∪ script-routed PII encoders, minus medicine and lab-row spans. **(3) Flag**: rules compare each value with the report's own range. **(4) Explain**: a Bengali template by default; the optional English LLM sees only the fact sheet and is shown only if the number + coverage gate passes. The Android app runs OCR, rules, templates and all three encoders on the phone (Sec. VI-D); the LLM was evaluated on a GPU.

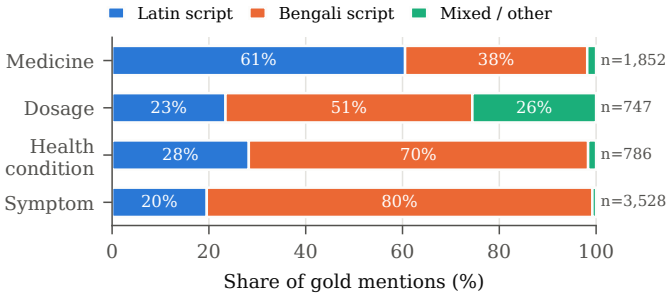


Fig. 2. Script of gold entity mentions in Bangla-HealthNER (train). Medicine names are mostly Latin-script, symptoms mostly Bengali-script; dosages often mix both (e.g., "1+0+1" followed by a Bengali "after meals").

TABLE I
EVALUATION SETS (ALL IDENTIFIERS ARE INVENTED).

Set	Docs	Used for
Bangla-HealthNER train/val/test	31,783	NER training and test (E0, E2, E4, E5)
Synthetic dev (vocabulary A)	300	threshold and setting selection
Synthetic held-out (vocabulary B)	300	de-identification, lab flags, explanation
Real text + injected identifiers	400	de-identification on real language
Free-text names (dev / test)	400/400	model-only privacy, quantization
Printed sheets	30	OCR (E8)

V. METHODS

A. Task and threat model

The input is a report x, either typed or read from a photo, in English, Bengali or code-mixed text. The system outputs (i) a de-identified text $\tilde{x}$ in which every personal identifier (name, phone, national ID, record ID, address, facility, date, doctor) is replaced by a typed tag, (ii) a list of lab results $\{(t_i, v_i, u_i, r_i, f_i)\}$ with test name, value, unit, reference range and flag $f_i \in \{\text{high}, \text{low}, \text{normal}\}$, and (iii) a plain-language explanation y in the user's language. Processing is local, so the main privacy risk is not the network but what leaves the pipeline afterwards: screenshots and shared summaries, and any optional larger model the user may choose to consult. De-identification therefore runs before any text is passed to a generator, and the generator receives only the fact sheet. Two errors matter and they trade off: a *leak* (an identifier survives) and *clinical over-redaction* (a medicine or lab value the patient needs is hidden). For explanations, a *wrong-direction* error (calling a high value low) is the most harmful failure, followed by *misleading interpretation* (causes, diagnoses or advice not supported by the report).

B. Hybrid de-identification with clinical protection

Rules. Format rules detect Bangladeshi mobile numbers (with Bengali digits), 10/13/17-digit national IDs, record identifiers, facility names (name words plus a suffix such as "Diagnostic Centre"), addresses with a gazetteer of the 64 districts, and labelled address fields. A second rule set (v2), added for the mobile version, covers numeric and textual dates, name fields ("Patient:", the Bengali "patient's name:") and the doctor title ("Dr.", Bengali "Ḍā.").

Models. OpenMed PII models [6] are routed by script: the English model for Latin-only text, the Bengali model for Bengali-only text, and the union of both for mixed text. The output is the union of rule and model spans.

Protection. The fine-tuned medicine tagger (Sec. V-C) marks Medicine and Dosage spans, and the lab parser marks lab rows. These spans are subtracted from model spans (never from rule spans), so a PII model cannot erase "Napa 500" or "SGPT (ALT) 1 U/L 7–45". Thresholds are chosen on the development set as the setting with the lowest leak whose clinical over-redaction is at most 10%.

Metrics. A gold identifier *leaks* if any of its letters or digits remains visible. *Clinical over-redaction* is the share of gold medicine and lab-value spans that are partly hidden. We report bootstrap 95% confidence intervals over documents [24], McNemar tests on "document has any leak" [25], and, when no leak is observed, the rule-of-three upper bound $3/n$ [26].

C. Code-mixed medicine NER

We fine-tuned BanglaBERT, BanglishBERT [10], XLM-RoBERTa-base [11] and the OpenMed Bengali PII encoder on Bangla-HealthNER (BIO tags on the first sub-word of each word; 4 epochs, batch 32, learning rate 3×10^{-5}, best epoch on validation; three seeds for the top models), plus LoRA ($r = 16$) variants [27]. We score exact-span micro-F1 and Medicine recall separately for Bengali-script and Latin mentions. The chosen model was exported to ONNX and dynamically quantized to INT8 [20], [28].

D. Rule-grounded explanation with small LLMs

Lab rows and flags. A parser extracts "test value unit range" rows, accepting Bengali digits, decimal commas and one-sided ranges ("$<$ 4.5", "$>$ 30"), and flags each value against the report's own range, with range ends counted as normal. Using the printed range rather than a population table keeps the decision tied to the laboratory's own method and units.

Template. For each row, the template states the test, the value with its unit, and the status with the range; Bengali output uses a patient-facing Bengali name followed by the original English name (e.g., "creatinine (kidney test)"), so that the patient can match it to the paper. It ends with a fixed sentence recommending a doctor visit. Because every sentence is assembled from parsed fields, the template cannot invent numbers or reverse a flag.

LLM conditions. We compared three prompt conditions with Qwen2.5-0.5B/1.5B-Instruct [16], Qwen3-1.7B [17] and Phi-3.5-mini (3.8B) [18], loaded in 4-bit NF4 [19] with greedy decoding: *free* (the model reads the de-identified report), *grounded* (the model rewrites the rule-computed fact sheet) and *strict* (grounded, plus explicit rules: one sentence per test, no causes, diseases, food, medicines or numbers beyond the facts). A deterministic bilingual *template* with patient-facing Bengali test names is the baseline. Each output passes a *gate*: a number check rejects any number not in the facts (Bengali digits normalized), and a coverage check rejects outputs that omit an abnormal test or its value; rejected outputs fall back to the template (Appendix A). We ran 100 held-out reports $\times$ {EN, BN} per model and condition, with 320 and 1,024 new tokens.

LLM judge. Qwen2.5-14B-Instruct (4-bit) judged each output on correctness, safety (no misleading interpretation, diagnosis or treatment advice) and clarity, using axes adapted from Med-PaLM [13] and PEMAT [14]. A first, holistic rubric failed validation (Cohen's $\kappa = 0.00$ for correctness [29]) because the judge marked fully correct outputs as wrong. The accepted rubric (v2) asks the judge only to *extract* the status stated for each test, without the answer key; code then scores correctness against the rule flags. Validation used 60 outputs rated independently by a separate commercial LLM (Claude), which we disclose as a limitation.

Native-speaker check. A native Bengali speaker (the author) rated 30 reports $\times$ {template, best LLM} in random order without system labels for status correctness, clarity (1–5), naturalness (1–5) and safety.

Algorithm 1 ShasthoSathi inference on one report

Require: image I or text x; UI language $\ell \in \{\text{en}, \text{bn}\}$; PII threshold τ
Ensure: de-identified text $\tilde{x}$, lab facts F, explanation y

if input is an image I **then**
  $W \leftarrow \text{OCR}(I)$ ▷ ML Kit or Tesseract (Auto)
  $x \leftarrow \textsc{RebuildRows}(W)$ ▷ slope-corrected row grouping
  user confirms or edits x against I
$x \leftarrow \textsc{Normalize}(x)$ ▷ Bengali digits, whitespace
$S_R \leftarrow \textsc{RuleSpans}(x)$ ▷ phone, NID, IDs, dates, names
$S_M \leftarrow \emptyset$
for $m \in \textsc{Route}(x)$ **do** ▷ bn, en, or both by script
  $S_M \leftarrow S_M \cup \textsc{Group}(\textsc{Tag}_m(x), \tau)$
$P \leftarrow \text{NER}_{\text{med}}(x) \cup \textsc{LabRows}(x)$ ▷ protected spans
$S \leftarrow \textsc{Merge}\big(S_R \cup \textsc{Trim}(S_M \setminus P)\big)$
$\tilde{x} \leftarrow \textsc{Replace}(x, S)$ ▷ e.g. [NAME], [PHONE]
$F \leftarrow \{(t, v, u, r, \textsc{Flag}(v, r))\}$ from $\textsc{ParseLabs}(\tilde{x})$
$y \leftarrow \textsc{Template}(F, \ell)$
if $\ell = \text{en}$ **and** LLM enabled **then**
  $y' \leftarrow \text{LLM}(\textsc{StrictPrompt}(F))$ ▷ never sees x
  if $\textsc{Numbers}(y') \subseteq \textsc{Numbers}(F)$ **and** every abnormal $t \in F$ with its v appears in y' **then**
    $y \leftarrow y'$
return $\tilde{x}, F, y$

Algorithm 1 summarizes inference end to end. Every stage after OCR is deterministic except the optional LLM call, whose output is shown only if the gate accepts it.

E. Model compression for mobile devices

All three encoders are exported to ONNX and compressed with post-training *dynamic* quantization in ONNX Runtime [28]: weights are stored as signed 8-bit integers with a per-tensor (or per-channel) scale, and activations are quantized on the fly from their observed range at each call [20]. No calibration data or retraining is needed, which matters because no Bengali clinical calibration corpus with identifiers exists. We compare six variants of the PII models: FP32; full INT8 (all MatMul and Gather weights); *reduced range* (7-bit weights, which avoids overflow in 16-bit accumulation on CPUs without VNNI instructions); per-channel scales; *attention excluded* (the query–key and attention–value MatMuls stay in FP32); and *embedding-only* INT8, which quantizes only the Gather over the word-embedding table. Embedding-only quantization stores the largest tensor in 8 bits (the Bengali model has a 250k-token vocabulary) but performs no activation quantization at all, so every matrix product is computed exactly as in FP32. Variants are selected per language with Algorithm 2; the medicine tagger uses full INT8 because its F1 loss was 0.09 points.

F. Statistical analysis and implementation

Confidence intervals for leak rates are percentile bootstraps over documents (2,000 resamples) [24]; differences between de-identification systems use a paired bootstrap on per-document leak rates and McNemar's test on whether a document leaks at all [25]. NER models are compared with Welch's t-test over three seeds. Judge agreement is Cohen's κ [29], and the paired native-speaker comparison uses a

Algorithm 2 Per-language selection of a quantized PII model

Require: FP32 models $M_{\text{bn}}, M_{\text{en}}$; variants $\mathcal{V}$; aggregations $\mathcal{A} = \{\text{first}, \text{mean}, \max\}$; thresholds $\mathcal{T}$; dev sets D (labelled) and D_{free} (unlabelled names); limits $c_{\max} = 0.10$, $o_{\max} = 0.10$, $B = 700\,\text{MB}$

for $\ell \in \{\text{bn}, \text{en}\}$ **do**
  for $v \in \mathcal{V}$ **do**
    $Q_{\ell,v} \leftarrow \textsc{QuantizeDynamic}(M_\ell, v)$
    check tokenizer parity and $\max |z_Q - z_{\text{FP32}}|$ on dev words
$\mathcal{C} \leftarrow \emptyset$
for $(v_{\text{bn}}, v_{\text{en}}, a, \tau) \in \mathcal{V}^2 \times \mathcal{A} \times \mathcal{T}$ **do**
  run Alg. 1 privacy stage on D and, models only, on D_{free}
  $\lambda \leftarrow$ free-text names missed; $c \leftarrow$ clinical over-redaction; $o \leftarrow$ other text hidden
  if $c \le c_{\max}$ **and** $o \le o_{\max}$ **and** $|Q_{\text{bn},v_{\text{bn}}}| + |Q_{\text{en},v_{\text{en}}}| \le B$ **then**
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{(\lambda, \text{leak}_D, o, \cdot)\}$
$(v^*_{\text{bn}}, v^*_{\text{en}}, a^*, \tau^*) \leftarrow \arg\min_{\mathcal{C}} (\lambda, \text{leak}_D, o)$ ▷ lexicographic
on the phone, rerun the test sets; if its leak differs from the server's, replace v^* by an embedding-only variant ▷ E7d/E7e
return $(v^*_{\text{bn}}, v^*_{\text{en}}, a^*, \tau^*)$

two-sided sign test. All selection (thresholds, quantization variant, aggregation, size budget) uses the development sets only; test sets are scored once per configuration. Training and LLM experiments ran as batch jobs on NVIDIA H100 GPUs (PyTorch 2.14, Transformers 4.57, OpenMed 2.5); CPU latency was measured with ONNX Runtime 1.25 at batch size 1 with four threads, the setting closest to a phone.

The Android app separates a pure-Kotlin *core* (rules, parser, templates, gate, redaction, word-level decoding of model outputs, OCR row rebuilding and OCR metrics) from the user interface. The core has no Android dependency, so it is unit-tested on any computer against *golden files* produced by the Python reference: for every test text, the files store the expected rule spans, parsed rows, redacted text and template output, and a test fails on any character difference. Regular expressions and Bengali text resources are generated from the Python sources rather than retyped.

VI. Results

A. De-identification (E1, E3, E3b)

Fig. 3 summarizes the three components; the subsections below give the details. Table II summarizes privacy. OpenMed models alone leaked 12.2% of identifier spans on the development set (facility names 41%, record IDs 45%) and 10.3% on the held-out set. Model errors concentrated in types with a clear format: facility names and record IDs, which rules catch completely. Rules alone, in turn, missed every name and date (53.6%), so the two sources are complementary. The English model applied to Bengali notes leaked 53–71% of spans, against 11–24% for the Bengali model, which motivates routing by script. Their union cut leaks to 0.6% on held-out and 0.2% on real text (McNemar 148 vs. 0 and 107 vs. 0 discordant documents). However, on real Bengali posts the PII models also erased 36.9% of gold medicine and dosage spans. Protection with the fine-tuned tagger reduced this to 6.2% at the same threshold without adding any leak, which in turn allowed a lower threshold (0.3) chosen on the development set: no leak was observed on either test set (rule-of-three upper bounds 0.16% and 0.15%) with 6.9% clinical over-redaction on real text. Fig. 4 shows the trade-off: lowering the threshold of the PII models alone trades leak for over-redaction along one curve, whereas rules and protection move the system off that curve to the lower-left corner.

TABLE II
De-identification: identifier leak and clinical over-redaction.

	Held-out synthetic		Real text + injected	
System (thr.)	Leak [CI]	Over	Leak [CI]	Over
Rules only	53.6 [53.1,54.1]	0.0	60.0	0.0
OpenMed PII (0.5)	10.3 [9.0,11.6]	8.4	5.5 [4.7,6.5]	36.9
Hybrid (0.5)	0.6 [0.3,1.1]	8.4	0.2 [0.1,0.4]	36.9
+ protect (0.5)	0.6	**0.0**	0.2	6.2
+ protect (0.3)	**0.0** [0,0]	0.1	**0.0** [0,0]	6.9

Values in %; CI = bootstrap 95%. Over = clinical over-redaction (gold medicine/lab spans partly hidden). "+ protect" = hybrid with medicine/dosage protection. Thresholds chosen on the development set.

B. Code-mixed medicine NER (E2, E4, E5)

Zero-shot OpenMed clinical NER found 70–75% of Latin-script medicines in Bangla-HealthNER but 0–9% of Bengali-script ones (E2, lenient overlap matching). Fine-tuning closed this gap (Fig. 5): Bengali-script Medicine recall reached 81–82%. The three full fine-tuned models were within one standard deviation of each other (Welch $t \approx 1.8$, not significant) and 1.7–2.6 points above the published 56.13 [5]; we attribute part of the gap to setup differences and claim parity, not a new state of the art. LoRA lost 7–9 points. BanglishBERT, with 40% of the parameters and the lowest variance, was exported: INT8 ONNX is 4× smaller and 2.7× faster than PyTorch on CPU with a 0.09-point F1 loss and 99.1% of word tags unchanged. The confusion matrix of the shipped tagger (Fig. 6) shows where the remaining errors lie: Medicine (93% of words), Specialist (95%) and Age (87%) are tagged reliably, and errors are mostly *missed* words rather than wrong types: 26% of Symptom and 32% of Health Condition words were predicted as ○. The one substantial type confusion is Health Condition read as Symptom (17%), two categories whose boundary is often unclear in lay descriptions. Only 0.9% of Medicine words were given another entity type, so medicine protection fails mainly by omission, which the rule-based lab-row protection partly covers. Entity-level F1 on the same run is 82.5 for Medicine, 45.4 for Dosage and 42.0 for Symptom (micro 57.3), matching the INT8 result in Table VI.

C. Explanation quality (E6, E6c, E6d)

LLMs must not decide abnormality. Reading reports directly, every model stated the wrong direction (high vs. low) for 17–42% of abnormal results (keyword proxy). The rule-computed fact sheet removed nearly all direction errors and raised judged English correctness from 4–55% to 22–98% (Fig. 7; exact values in Table VII, Appendix B).

Correct is not safe. With the fact sheet, the best models still added a misleading interpretation in about a third of outputs (e.g., "low HbA1c means better sugar control"), and one model recommended supplements. The strict prompt cut misleading

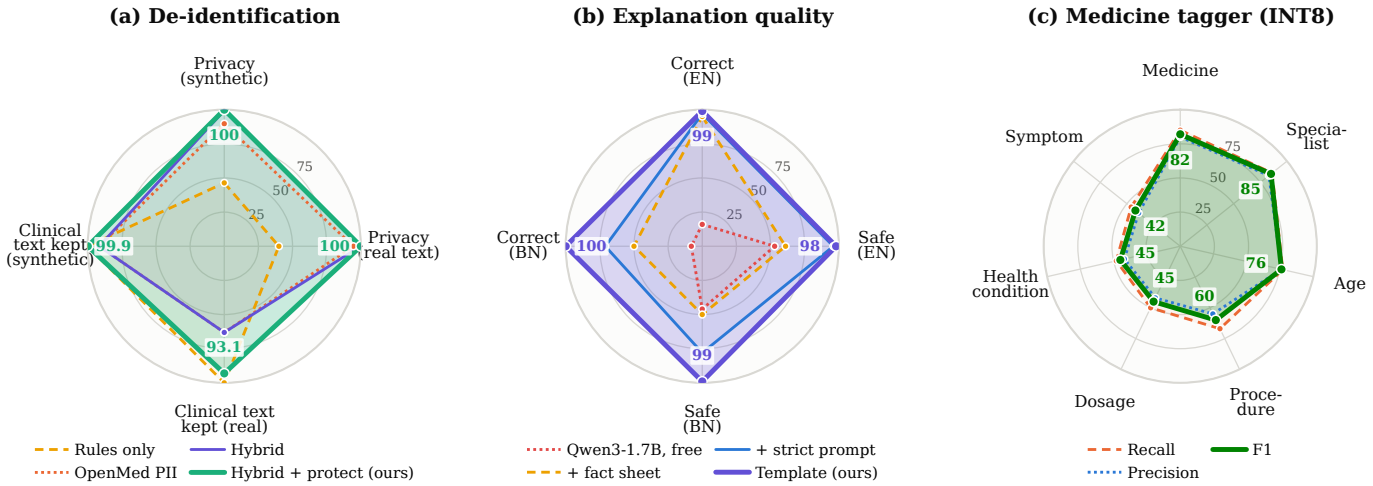


Fig. 3. Results at a glance (all axes 0–100, higher is better; bold numbers label the shipped configuration). **(a)** De-identification: privacy = 100−identifier leak and clinical text kept = 100−over-redaction, on held-out synthetic reports and real text with injected identifiers (Table II). **(b)** Judged explanation quality for Qwen3-1.7B as grounding increases, against the deterministic template (Table VII). **(c)** Per-type precision, recall and F1 of the shipped INT8 medicine tagger on the Bangla-HealthNER test split (word level, Fig. 6).

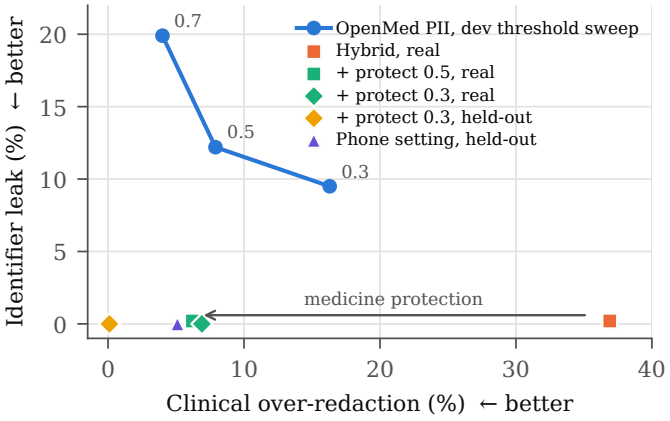


Fig. 4. Privacy–utility trade-off. The blue curve is the OpenMed PII models alone at thresholds 0.3/0.5/0.7 (development set); markers are full systems. Medicine protection moves the hybrid system from 36.9% to about 6% clinical over-redaction on real text with no added leak.

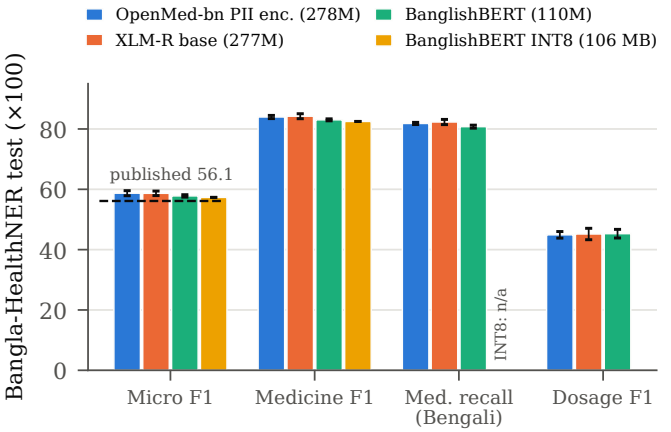


Fig. 5. Bangla-HealthNER test results (exact span, ×100; mean ± SD over three seeds; INT8 one run). Dashed line: published BanglishBERT micro-F1 [5]. Exact values, BanglaBERT and LoRA results are in Table VI (Appendix B).

claims for Qwen3-1.7B from 38 to 4 per 100 outputs and raised safety from 60% to 96% with no loss of correctness. The number check cannot catch a report's own number used in the wrong role (Table III); the strict prompt and the coverage check reduce, but do not remove, this risk. Its English outputs passed the number and coverage gate on 8 of 8 demonstration reports, but were then close to a copy of the fact sheet: once constrained enough to be safe, the LLM adds little over the template.

Bengali fails. No model produced usable Bengali. Under the free and grounded conditions, 55–99% of abnormal results went unmentioned and up to 77% of outputs looped. Tokenization explains part of this. Bengali consumed about one token per character, versus roughly 3.5 characters per token for English, so at the same 320-token limit the median Bengali answer was 305 characters against 909 for English, consistent with tokenizer disparity across languages [4]. Raising the limit to 1,024 tokens ruled out truncation as the cause: English results were unchanged, while 66–100% of Bengali outputs now looped, so the failure lies in the models' Bengali generation rather than in the budget. The strict prompt improved Qwen3 to 71% correct by the judge, but the native-speaker check (Table IV) rated it 60% correct, 2.27/5 clear and 1.03/5 natural, against 100%, 5.00 and 5.00 for the template, which won on all 30 pairs (sign test $p < 0.001$). The judge agreed with the native rater on 82% of correctness and 87% of safety decisions but over-rated Bengali clarity (4.21 vs. 2.27); we therefore use human ratings only for Bengali quality.

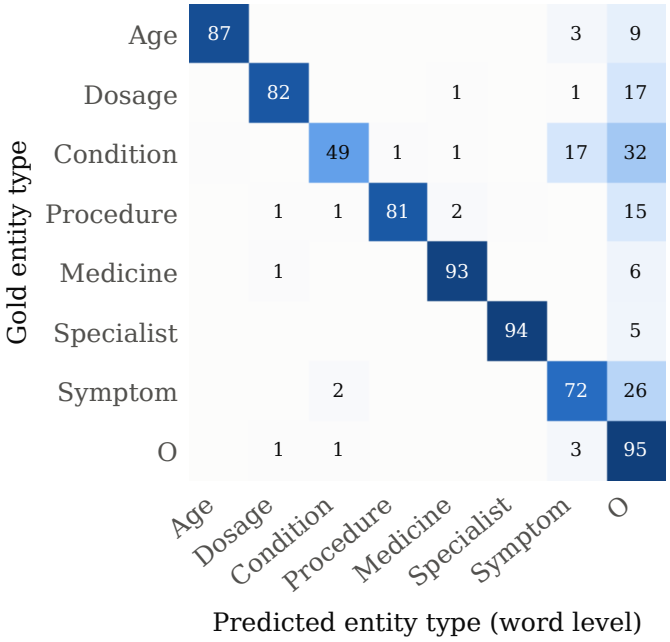


Fig. 6. Word-level confusion matrix of the shipped INT8 medicine tagger on the Bangla-HealthNER test split (rows: gold type, row-normalized %; B/I collapsed). Off-diagonal mass in the O column is missed words; in the O row, spurious ones. 3,179 posts; INT8, first sub-word decoding.

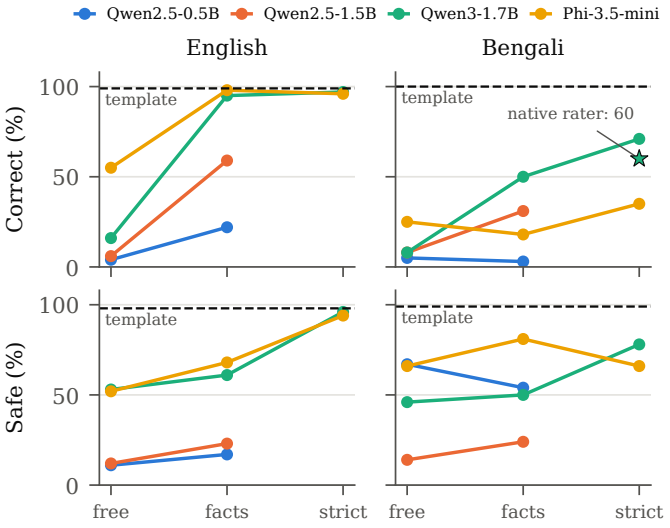


Fig. 7. Judged correctness and safety of four small LLMs as grounding increases (free: reads the report; facts: rewrites the rule-computed fact sheet; strict: facts plus explicit constraints; 100 held-out reports each). Dashed: deterministic template. Star: blinded native-speaker correctness for Qwen3-1.7B strict in Bengali.

D. Mobile deployment (E7)

Export. The tagger and both PII models were exported to ONNX with their tokenizer files; tokenizer outputs matched the reference on every test word and FP32 logits matched PyTorch (max. difference $< 10^{-5}$). A dependency-free reference implementation (onnxruntime and tokenizers only, word-level decoding) defines exactly what the phone computes.

TABLE III
FAILURE TYPES OBSERVED IN SMALL-LLM EXPLANATIONS (E6 AND E6C).

Failure	Example (English gloss for Bengali)
Wrong direction	A low haemoglobin described as high (free condition, 17–42% of abnormal results).
Unsupported inference	"Low HbA1c means better sugar control"; supplement advice.
Number in wrong role	"Haemoglobin is usually between 10.2 and 15.5", where 10.2 is the patient's value.
Looping (BN)	One word repeated until the token limit (up to 100% of outputs).
Wrong term (BN)	"Hematocrit" written for haemoglobin; a non-word used for "high".
Untranslated copy (BN)	The English fact sheet copied with Bengali connectives.

TABLE IV
BLINDED NATIVE-SPEAKER CHECK OF BENGALI OUTPUT (30 REPORTS).

System	Correct	Clear (1–5)	Natural (1–5)	Safe
Template	**100%**	**5.00**	**5.00**	100%
Qwen3-1.7B, strict	60%	2.27	1.03	100%

Quantization fragility. With plain dynamic INT8 PII models, the leak on held-out reports rose from 0% (research pipeline) to 16.5%. Format rules hid this on labelled test sets, so we used the free-text set, where only the model can find a name. Figure 8 separates the two languages. The English model tolerated INT8 (1.5% of names missed, equal to FP32), but the Bengali model missed every unlabelled Bengali name. Range-reduced and attention-excluded variants only partly recovered it (14–26%), and per-channel variants predicted nothing. Quantizing *only the embedding table*, which accounts for most of the 250k-vocabulary model's size but involves no activation quantization, matched FP32 exactly (0% missed) at half the size. The configuration chosen on development data (Bengali embedding-only INT8, English INT8 except attention, mean word aggregation, threshold 0.1, rules v1+v2; 689 MB) gave 0.0% leak on held-out and real text, 1.0% on free text (Bengali 0%), 5.1% non-identifier text hidden (research pipeline 8.2%), no lab or medicine damage, and 181 ms per report on four server CPU threads. The 700 MB size budget was fixed after an earlier tuning round, and an earlier round that selected settings hiding 58% of all text was rejected; both are reported for transparency.

Hardware dependence (E7d, E7e). Rerunning the 1,100 test documents on the phone (see the Android app below) exposed a failure the server could not show. With the chosen English model (INT8 except attention), the phone missed names in 15.5% of free-text documents versus 1.0% on the server (62 vs. 4 documents; all 58 phone-only leaks were English names inside sentences), while held-out and real text stayed at 0%. The tokenizer was identical on both machines, and the Bengali embedding-only model gave identical word labels, which points to the INT8 matrix multiplications: ONNX Runtime executes them with different integer kernels on ARM

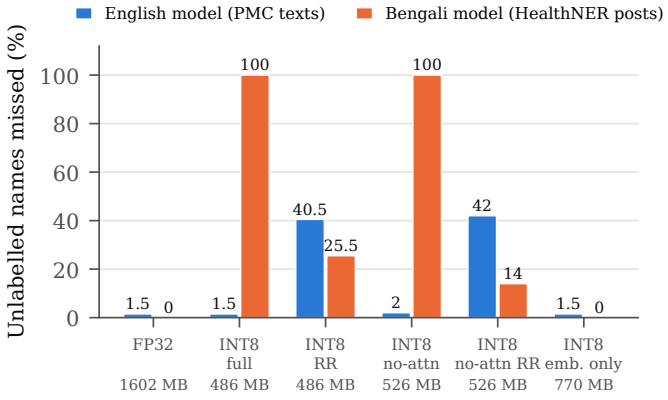
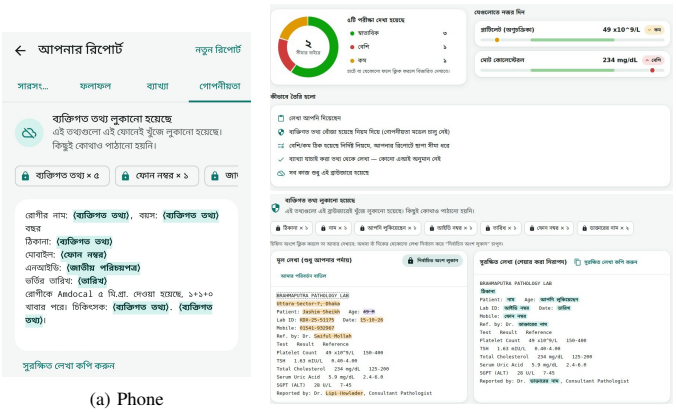


Fig. 8. Quantization fragility: unlabelled names missed on the free-text test set by the PII models alone (no rules), best setting per variant chosen on development data; model size below each group. RR = reduced range; no-attn = attention MatMuls kept in FP32; emb. only = only the embedding table quantized (Sec. V-E).



(a) Phone

(b) Web app

Fig. 9. ShasthoSathi in Bengali on invented sample reports. (a) Galaxy S23 Ultra, privacy view with the on-phone models active: name, age, address, phone, national ID, date and doctor are replaced by tags, while the medicine and dose ("Amdocal 5 mg, 1+1+0") stay visible. (b) Web app: summary with flagged results (top) and the side-by-side privacy view, in which the user can show or hide any span (bottom).

and x86. We therefore ship embedding-only INT8 for both PII models (770 MB, 70 MB over the budget; Table IX). On the server this configuration gave 0.0% leak on held-out and real text and 0.75% on free text, with 0–3.8% of medicine and lab spans and 1.7–6.2% of other text hidden.

Android app. The app is written in Kotlin with Jetpack Compose; a report flows from input and a mandatory text check to Summary, Results, Explanation and Privacy views, in separate Bengali and English interfaces (Fig. 9). Regular expressions and text resources are generated from the Python sources, with each Python shorthand class rewritten into an explicit Unicode class, because Java, Android (ICU) and Python disagree on what \w, \s and \b match. On 909 texts (600 synthetic reports, 300 Bangla-HealthNER posts and edge cases), the Kotlin core reproduced the Python reference exactly for rule spans, lab parsing, flags, templates, the gate, redaction and model-output decoding. The same check runs on the phone as a built-in self-test. On a Samsung Galaxy S23 Ultra (SM-S918B, Android 16, 8 cores) the self-test reported all six checks identical to Python (core, gate, tagger windowing, span grouping, OCR metrics and row rebuilding), and the rules-and-template pipeline took a median of 0.96 ms per report (p90 1.37 ms; 360 runs over 9 reports × 2 languages) with 13 MB of Java heap in use. In app v0.4 the three encoders also run on the phone, with ONNX Runtime for Android [28] and the Hugging Face tokenizer library through the Deep Java Library bindings [30] to the Rust implementation used in Python. The user copies the model folder (900 MB, including the server's configuration file) to the phone once and selects it with the system folder picker; the app copies it into private storage and needs no storage or network permission. Against golden files generated on the server, the phone produced identical token ids for all three models and identical word labels for both PII models (2,226 words; largest probability difference 3×10^{-5}), while the INT8 medicine tagger labelled 1.1% of words differently. A built-in leak test runs the E7 test sets (300 held-out, 400 real and 400 free-text documents) through the phone pipeline and scores them with the server's metric. The phone matched the server on every set (leak 0.0%, 0.0% and 0.75%; the same three free-text documents leaked on both) and hid identical spans in 1,097 of 1,100 documents; the other three differed only in medicine protection or other text, never in a leak. With all three models, the S23 loaded them in 3–4 s and processed a report in a median of 262 ms (p90 524 ms; 90 runs, four threads), of which the models took 260 ms; the app's total memory was 2.0 GB (proportional set size), almost all in the native runtime. The S23 is a flagship; this footprint is too large for low-cost phones. **Web app.** The same core was ported to JavaScript (regular expressions generated from the Python sources) and runs entirely in the browser with Tesseract.js OCR, optional ONNX Runtime Web models, a passphrase-encrypted, pseudonymised report history and a printable doctor-visit summary; a content-security policy restricts the page to its own origin. It reproduced the Python reference on all 2,327 golden cases in Node and in Chromium.

E. Reading photographed reports (E8)

The app reads photos on the device with ML Kit (Latin) [22] or Tesseract with Bengali and English models [21]; in *Auto* mode it keeps the Tesseract text when at least 10% of its letters are Bengali. OCR engines often return a lab table column by column, which breaks the row structure the parser needs. The app therefore rebuilds rows from word boxes: it estimates the page slope k as the median slope of the engine's own text lines, maps each word centre (c_x, c_y) to $c_y - k\,c_x$, groups words whose corrected heights differ by less than $0.55h$ (h the median word height), and orders each group left to right. On 60 reports cut into word boxes, tilted by 3° and shuffled column-wise, this recovered every row and every lab flag. The user must then compare the text with the photo before an explanation is produced, because a single misread digit can change a flag. In a desktop proxy study with the

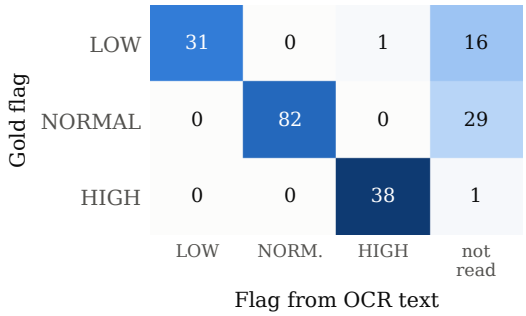


Fig. 10. Lab-flag confusion after OCR (desktop proxy, ben+eng, clean/tilted/half-resolution pages; counts). "not read": the row was not parsed from the OCR text.

TABLE V
OCR OF THE 30 PRINTED SHEETS: DESKTOP PROXY WITH THE PHONE'S TESSERACT MODELS (FAST BEN+ENG).

Setting	Condition	CER	Rows exact	Flags ok	Leak (rules)
Proxy	clean	1.6	65.2	81.8	8.4
Proxy	tilted 3°	2.5	48.5	71.2	15.3
Proxy	half resolution	2.7	42.4	75.8	12.4

Values in %. Leak is token-level and counts a shared surname visible elsewhere as a leak.

same Tesseract models on the 30 printed sheets (Table V), clean pages had a 1.6% character error rate, yet only 65% of lab rows were read exactly, because single-character errors hit units ("x10^9/L"→"x1049/L") and test names ("25-OH"→"25-0H"). OCR errors also defeated format rules ("Dr."→"pr.", a dropped leading zero in a phone number, a doctor's name wrapped onto the next line), raising the rules-only leak to 8–15%. An English-only OCR model turned Bengali notes into noise (77% character error rate). Fig. 10 shows that OCR errors almost never *reverse* a flag (1 of 198 rows); they cause rows to be missed, which the app reports as "not read" rather than guessing.

Scoring photographs taken on the phone (three conditions per printed sheet) is supported by the app and the printable sheets are released, but this evaluation is left to future work; Table V therefore reports the desktop proxy only.

VII. Discussion

Where small LLMs help. In English, a 1.7B model rewriting a rule-computed fact sheet under strict constraints is correct and safe, but by then it closely paraphrases the template. Its main value is stylistic variety, which must always pass the gate. For Bengali, the evidence favors a template-first design: the template is correct by construction and was preferred on every report, whereas the best small model produced unnatural text that an LLM judge rated as fluent. Bengali generation quality should therefore be measured by native speakers, not by LLM judges.

Hidden failure modes. Four failures were invisible to the headline metrics and appeared only with targeted tests: PII models erasing medicines on real Bengali text, INT8 quantization disabling the Bengali PII model, an INT8 English model that worked on the server but missed names on the phone, and OCR errors breaking format rules. Three lessons follow. Synthetic test sets whose identifiers follow field labels let rules look perfect, so model-only tests on unlabelled text are needed. And quantization must be validated per language and per task, not by average agreement with the FP32 model. Finally, a quantized model must be re-validated on the target hardware: the same model file missed 15 times more names on the phone.

Implications for low-resource clinical NLP. The division of labour that worked here (encoders for recognition, rules for decisions, templates for language, and LLMs only where they are verifiably safe) is cheap, auditable and portable to other languages in which small LLMs are weak. Its cost is coverage: a template can only say what its designers anticipated, and the parser handles tabular lab rows, not narrative findings. Closing that gap safely likely requires Bengali-specific small models with efficient tokenizers, evaluated by native speakers and clinicians, rather than larger multilingual models judged by other LLMs.

Limitations. Identifiers and reports are synthetic or injected; real Bangladeshi reports, West Bengal and diaspora documents, and handwriting were not tested. The native-speaker check had one rater, who is also the author, and the template's fixed pattern makes blinding partial. The judge-validation reference was produced by another LLM, and no clinician reviewed the explanations. The lab parser covers tabular rows only, and the test vocabulary of the synthetic generator is small. The v2 header rules were designed for the mobile version after the earlier experiments, phone results come from one flagship device, and the shipped privacy models (770 MB) exceed our 700 MB budget, with the app using 2.0 GB of memory, too much for low-cost phones; and OCR was evaluated on rendered pages, not phone photographs.

Ethics. No real patient data were used; all identifiers are invented. The app gives no diagnosis or treatment advice, always recommends showing the report to a doctor, and sends no data off the device.

VIII. Conclusion

A privacy-first pipeline built from small models can de-identify code-mixed Bengali health reports, keep the clinical content patients need, and explain laboratory results correctly, provided that rules, not LLMs, decide abnormality. Small LLMs are useful English rewriters when strictly grounded and gated, but in Bengali a deterministic template was safer, clearer and more natural than every model we tested. The whole pipeline runs offline on a phone, but its privacy matched the server only after the quantized models were re-validated on the phone itself. Future work includes multi-rater Bengali evaluation with clinicians, real-report and phone-photograph OCR studies on low-cost phones, smaller on-device models for low-cost phones, and Bengali-specific small models with efficient tokenizers.

Code and Data Availability

Project page with all results, figures and a live demo: https://cimilab.github.io/Bangla_ShasthoSathi/. In-browser web app: https://cimilab.github.io/Bangla_ShasthoSathi/app/. Source of the page and app: https://github.com/CIMILab/Bangla_ShasthoSathi. Code, Slurm scripts, synthetic data generators, golden files and the Android project are also in the `final_project` folder of the course GitLab repository.

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Appendix A
Number and coverage gate

The gate that decides whether an LLM explanation may be shown (Kotlin port; the Python version is identical in behaviour and golden-tested):

```
fun gate(text: String, labs: List<LabResult>,
         lang: String): Gated {
  val clean = GATE_BULLET.replace(
      text.replace("**", ""), "")
  val bad = Rules.unsupportedNumbers(clean, facts(labs))
  val low = clean.lowercase()
  val missing = labs.filter { l ->
    (l.flag == "high" || l.flag == "low") &&
    (NAME_PAREN.replace(l.name, "").lowercase()
       .replace("s. ", "") !in low ||
     Py.g(l.value) !in clean) }.map { it.name }
  return if (bad.isNotEmpty() || missing.isNotEmpty())
    Gated("template", template(labs, lang), bad, missing)
  else Gated("llm", text, emptyList(), emptyList())
}
```

TABLE VI
BANGLA-HEALTHNER TEST (EXACT-SPAN, $\times 100$; MEAN $\pm$ SD OVER THREE SEEDS).

Model (params)	Micro F1	Medicine F1	Med. R (BN)	Dosage F1
OpenMed-bn PII enc. (278M)	**58.71**$\pm$0.84	83.97$\pm$0.53	81.83$\pm$0.36	44.91$\pm$1.09
XLM-R base (277M)	58.67$\pm$0.75	**84.23**$\pm$0.83	**82.31**$\pm$0.86	45.18$\pm$1.89
BanglishBERT (110M)	57.81$\pm$0.40	83.03$\pm$0.33	80.77$\pm$0.50	**45.29**$\pm$1.43
BanglaBERT (110M)[†]	55.75	81.83	83.36	43.13
XLM-R, LoRA r=16[†]	49.93	78.54	75.61	36.82
BanglishBERT INT8 ONNX[†]	57.32	82.50	–	–

[†]One seed. INT8: 106 MB, 19.5 ms per document on 4 CPU threads (PyTorch FP32: 420 MB, 53.2 ms).

TABLE VII
JUDGED EXPLANATION QUALITY (% OF 100 OUTPUTS, 1,024 TOKENS).

	English		Bengali	
Model / condition	Correct	Safe	Correct	Safe
Qwen2.5-0.5B: free $\rightarrow$ grounded	4 $\rightarrow$ 22	11 $\rightarrow$ 17	5 $\rightarrow$ 3	67 $\rightarrow$ 54
Qwen2.5-1.5B: free $\rightarrow$ grounded	6 $\rightarrow$ 59	12 $\rightarrow$ 23	8 $\rightarrow$ 31	14 $\rightarrow$ 24
Qwen3-1.7B: free $\rightarrow$ grounded	16 $\rightarrow$ 95	53 $\rightarrow$ 61	8 $\rightarrow$ 50	46 $\rightarrow$ 50
Phi-3.5-mini: free $\rightarrow$ grounded	55 $\rightarrow$ 98	52 $\rightarrow$ 68	25 $\rightarrow$ 18	66 $\rightarrow$ 81
Qwen3-1.7B, strict	**97**	**96**	71	78
Phi-3.5-mini, strict	96	94	35	66
Template (no LLM)	99	98	**100**	**99**

Judge v2: correctness $\kappa = 0.80$ (93% agreement), safety $\kappa = 0.66$ (83%), $n = 60$. The template is correct by construction; its 1–2% shortfall estimates judge error.

APPENDIX B
EXTENDED RESULTS

TABLE VIII
OPENMED PII MODELS ALONE ON THE DEVELOPMENT SET (E1).

Route (threshold)	Leak [95% CI]	Docs with leak	Over-red.
English model (0.5)	23.6 [21.3, 25.8]	79%	6.8
Bengali model (0.5)	28.8 [27.2, 30.5]	93%	14.1
By script (0.3)	9.5 [8.4, 10.7]	53%	16.3
By script (0.5)	12.2 [11.0, 13.4]	62%	7.9
By script (0.7)	19.9 [18.5, 21.2]	85%	4.0

TABLE IX
MOBILE PRIVACY TUNING (PII MODELS ALONE, SERVER; % OF UNLABELLED NAMES MISSED). THE E7C CHOICE FAILED ON THE PHONE (SEC. VI-D); EMBEDDING-ONLY FOR BOTH IS SHIPPED.

PII models	MB	Free dev	Free test	Free BN
FP32 (both)	1602	3.3	0.7	0.0
INT8 plain (both)	486	52.7	50.7	100.0
INT8 reduced range (both)	486	35.7	33.0	25.5
INT8 embedding only (both; shipped)	**770**	3.3	0.7	0.0
BN emb. only + EN no-attn (E7c)	689	3.5	1.0	0.0

APPENDIX C
REPRODUCIBILITY

Experiments: `scripts/05--17` (de-identification, NER, export, explanation, judge, native rating, mobile tuning, golden files, OCR); Slurm jobs in `slurm/`; Android project in `mobile/android`; printable E8 sheets in `mobile/eval`. Software: PyTorch 2.14, Transformers 4.57.6, OpenMed 2.5.0, ONNX Runtime 1.25, Tesseract 5.3.4 (desktop) and Tesseract4Android 4.9.0 (phone), NVIDIA H100 GPUs for training and LLM runs.